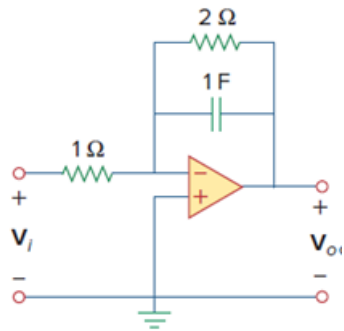


Problem

Scale the low-pass active filter in Fig. 14.99 so that its corner frequency increases from 1 rad/s to 200 rad/s. Use a 1- μ F capacitor.

Figure 14.99:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 14.99 in the textbook.

Write the expression for new capacitance.

$$C' = \frac{C}{K_m K_f} \dots\dots (1)$$

Here,

C is the old capacitance

K_m is the magnitude scale factor

K_f is the frequency scale factor

Calculate the frequency scale factor.

$$K_f = \frac{\omega'}{\omega}$$

Substitute 200 rad/s for ω' and 1 rad/s for ω .

$$\begin{aligned} K_f &= \frac{200}{1} \\ &= 200 \end{aligned}$$

Step 2 of 2

Recall equation.

$$C' = \frac{C}{K_m K_f}$$

Substitute 200 for K_f , 1 F for C and $1 \mu\text{F}$ for C' .

$$10^{-6} = \frac{1}{K_m (200)}$$

$$\begin{aligned} K_m &= \frac{1}{(10^{-6})(200)} \\ &= 5000 \end{aligned}$$

Calculate the new value of resistance R' .

$$R' = K_m R$$

Substitute 5000 for K_m and 1Ω for R .

$$\begin{aligned} R' &= (5000)(1) \\ &= 5 \text{ k}\Omega \end{aligned}$$

Calculate the new value of feedback resistance R'_f .

$$R'_f = K_m R_f$$

Substitute 5000 for K_m and 2Ω for R_f .

$$\begin{aligned} R'_f &= (5000)(2) \\ &= 10 \text{ k}\Omega \end{aligned}$$

Therefore, the new values of low-pass filter are,

$$\boxed{\begin{aligned} C' &= 1 \mu\text{F} \\ R' &= 5 \text{ k}\Omega \\ R'_f &= 10 \text{ k}\Omega \end{aligned}} .$$